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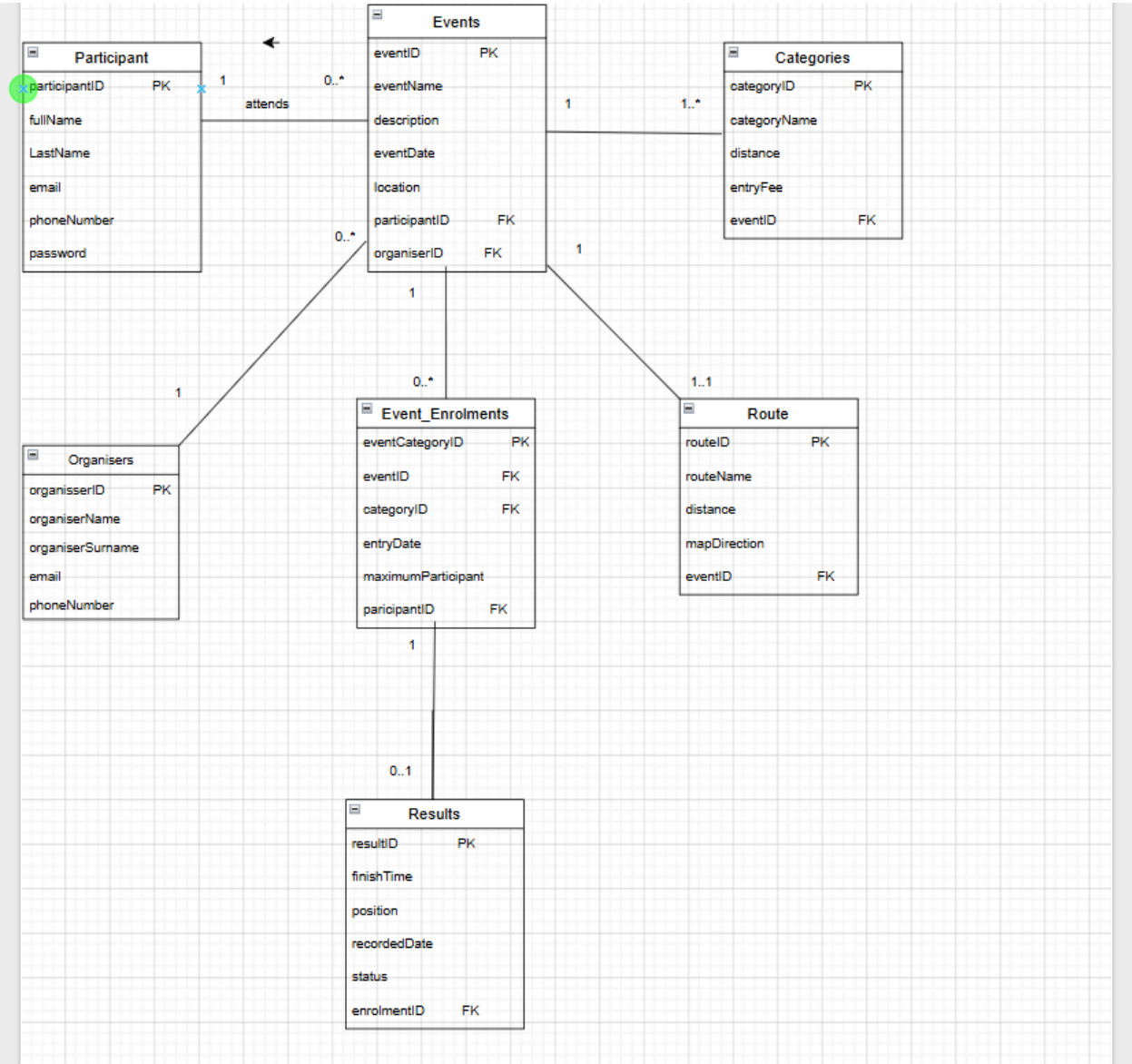
Module Name: Programming 2B

Module Code: Prog6212

ERD



SECTION B: API table

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
Post	/api/auth/register	Registers a new user as a Participant or Organizer	None (Public)	Full name, last name, email, password, phone number,role	201 Created – user registered successfully
Post	/api/auth/login	Allows users to log in according to their assigned role.	None (Public	Email, password	200 OK – Login successful and authentication token returned.

2. User profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Allows a user to view their own profile information.	Participant/ organiser	None	200 OK – User profile returned
PUT	/api/users/profile	Update the authenticated user's profile information	Any logged in users/ organizer	Updated profile information	200 OK – Profile updated successfully

3.Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Allows both roles to view available events.	Partcipant / Organisers	None	200 OK – List of events returned

GET	/api/events{id}	Allows both roles to view the details of a specific event.	Participant / Organser	None	200 Ok- Event details returned
POST	/api/evevents	Allows an organiser to create a new event.	Organiser	Event name, description, date, location, distance, and event type.	201 Created – Event created successfully
PUT	/api/events{Id}	Allows organiser to update an existing even	Oragniser	Updated event details	200 OK – Event Updated successfully
Delete	/api/event/{id}	Allows an organiser to delete an event.	Oragniser	None	204 No Content – Event deleted successfully

4.Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Allows users to view categories available for an event.	Participant/ Organiser	None	200 OK – Available categories returned
POST	/api/events/{eventId}/categories	Allows an organiser to define an age or distance category	Oragniser	Category name, age/distance and maximum	201 Created – Category created successfully

		for an event.		participants.	
PUT	/api/categories/{id}	Allows an organiser to update a category.	Oragniser	Updated category information.	200 OK – Category updated successfully
DELETE	/api/categories/{id}	Allows an organiser to delete a category.	Oragniser	None	204 No content- Category deleted successfully

5.Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments	Allows a participant to enter an event by selecting a category.	Participant	Selected category and payment status.	201 Created – Event enrolment created successfully.
GET	/api/enrolments/my	Allows an organiser to view all participants enrolled in their event.	Organiser	None	200 OK – List of event enrolments returned.
GET	/api/events/{eventId}/enrolments	Allows a participant to view their own event enrolment.	Participant	None	200 OK – Enrolment details returned.

DELETE	/api/events/{eventId}/enrolments	Allows a participant to cancel their event enrolment.	Participant	None	204 No Content – Enrolment cancelled successfully.

6.Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Allows an organiser to record a participant's finishing time and position.	Organiser	Finish time, finishing position, recorded date and status.	201 Created – Result recorded successfully.
GET	/api/results/my	Allows a participant to view their own race results.	Participant	None	200 OK – Participant's results returned.
GET	/api/events/{eventId}/results	Allows an organiser to view results for participants in their event.	Organiser	None	200 OK – Event results returned.
PUT	/api/results/{resultId}	Allows an organiser to update a participant's result.	Organiser	Updated finish time, position and status.	200 OK – Result updated successfully.

7. Routes

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/route	Allows users to view route information for an event.	Participant/ Organiser	None	200 OK – Route name, distance and map directions returned.
POST	/api/events/{eventId}/route	Allows an organiser to add route information to an event.	Organiser	Route name, distance and map directions.	201 Created – Route created successfully.
PUT	/api/routes/{routeId}	Allows an organiser to update route information.	Oragniser	Updated route information.	200 OK – Route updated successfully.

Section C: SQL

```sql
create database RaceDay;
use RaceDay;


CREATE TABLE Organisers (
    organisserID    INT IDENTITY(10,1) PRIMARY KEY,
    organiserName   VARCHAR(250)    NOT NULL,
    organiserSurname VARCHAR(250)    NOT NULL,
    email          VARCHAR(250)  NOT NULL UNIQUE,
    phoneNumber     VARCHAR(250)     NULL
);

CREATE TABLE Participant (
    participantID   INT IDENTITY(20,1) PRIMARY KEY,
    fullName        VARCHAR(250)     NOT NULL,
    LastName       VARCHAR(250)     NOT NULL,
    email          VARCHAR(250)   NOT NULL UNIQUE,
    phoneNumber     VARCHAR(250)    NULL,
    password        VARCHAR(250)   NOT NULL
);
);
CREATE TABLE Events (
    eventID          INT IDENTITY(30,1) PRIMARY KEY,
    eventName        VARCHAR(200)   NOT NULL,
    description      VARCHAR(255)   NULL,
    eventDate        DATE            NOT NULL,
    location         VARCHAR(250)   NOT NULL,
    participantID    INT             NULL,
    organiserID      INT             NOT NULL,

        FOREIGN KEY (participantID) REFERENCES Participant(participantID),
    CONSTRAINT FK_Events_Organisers
        FOREIGN KEY (organiserID) REFERENCES Organisers(organisserID)
);

CREATE TABLE Categories (
    categoryID      INT IDENTITY(1,1) PRIMARY KEY,
    categoryName    VARCHAR(50)    NOT NULL,
    distance        DECIMAL(6,2)    NOT NULL,
    entryFee        DECIMAL(8,2)    NOT NULL DEFAULT 0,
    eventID         INT             NOT NULL,

        FOREIGN KEY (eventID) REFERENCES Events(eventID)
);

CREATE TABLE Route (
    routeID         INT IDENTITY(60,1) PRIMARY KEY,
    routeName       VARCHAR(100)   NOT NULL,
    distance        DECIMAL(6,2)    NOT NULL,
    mapDirection    VARCHAR(255)   NULL,
    eventID         INT             NOT NULL UNIQUE,

        FOREIGN KEY (eventID) REFERENCES Events(eventID)
);
```

```sql
CREATE TABLE Event_Enrolments (
    eventCategoryID     INT IDENTITY(100,1) PRIMARY KEY,
    eventID             INT             NOT NULL,
    categoryID          INT             NOT NULL,
    entryDate           DATE            NOT NULL DEFAULT GETDATE(),
    maximumParticipant  INT             NOT NULL,
    paricipantID        INT             NOT NULL,

        FOREIGN KEY (eventID) REFERENCES Events(eventID),

        FOREIGN KEY (categoryID) REFERENCES Categories(categoryID),

        FOREIGN KEY (paricipantID) REFERENCES Participant(participantID),

        UNIQUE (paricipantID, categoryID)
);
```

```sql
CREATE TABLE Results (
    resultID        INT IDENTITY(1,1) PRIMARY KEY,
    finishTime      TIME            NULL,
    position        INT             NULL,
    recordedDate    DATE            NULL,
    status          VARCHAR(20)    NOT NULL
                        CONSTRAINT CHK_Results_Status
                        CHECK (status IN ('Finished','DNF','DNS','DSQ')), enrolmentID  INT    NOT NULL UNIQUE,
    CONSTRAINT FK_Results_Enrolments
        FOREIGN KEY (enrolmentID) REFERENCES Event_Enrolments(eventCategoryID)
);
```

```sql
-- Organisers (2)
INSERT INTO Organisers (organiserName, organiserSurname, email, phoneNumber) VALUES
('Thabo', 'Mokoena', 'thabo.mokoena@raceday.co.za', '0821234567'),
('Sarah', 'van der Merwe', 'sarah.vdm@raceday.co.za', '0827654321');

-- Participants (2)
INSERT INTO Participant (fullName, LastName, email, phoneNumber, password) VALUES
('Lindiwe', 'Nkosi', 'lindiwe.nkosi@gmail.com', '0839876543', 'HASHED_PW_1'),
('James', 'Botha', 'james.botha@gmail.com', '0845551234', 'HASHED_PW_2');

-- Events (3) - organiserID 1 and 2 from above
INSERT INTO Events (eventName, description, eventDate, location, participantID, organiserID) VALUES
('Benoni Park Run Challenge', 'Annual community road running event', '2026-10-10', 'Benoni, Gauteng', 1, 1),
('Pretoria Cycle Classic', 'Charity cycling event through the city', '2026-11-15', 'Pretoria, Gauteng', 2, 2),
('Soweto Heritage Walk', 'Community fun walk celebrating local heritage', '2026-12-05', 'Soweto, Gauteng', NULL, 1);

-- Routes (one per event, 1..1)
INSERT INTO Route (routeName, distance, mapDirection, eventID) VALUES
('Benoni Main Loop', 10.00, 'https://maps.raceday.co.za/benoni-loop', 1),
('Pretoria City Circuit', 42.00, 'https://maps.raceday.co.za/pretoria-circuit', 2),
('Soweto Heritage Trail', 5.00, 'https://maps.raceday.co.za/soweto-trail', 3);
```

```sql
INSERT INTO Categories (categoryName, distance, entryFee, eventID) VALUES
('10km Run', 10.00, 150.00, 1),
('5km Fun Walk', 5.00, 80.00, 1),
('42km Cycle', 42.00, 300.00, 2),
('5km Fun Walk', 5.00, 80.00, 3),
('21km Half Marathon', 21.10, 200.00, 3);
INSERT INTO Event_Enrolments (eventID, categoryID, entryDate, maximumParticipant, paricipantID) VALUES
(1, 1, '2026-09-05', 500, 1),
(1, 2, '2026-09-06', 300, 2),
(2, 3, '2026-09-10', 250, 1),
(3, 5, '2026-09-12', 200, 2);


INSERT INTO Results (finishTime, position, recordedDate, status, enrolmentID) VALUES
('00:52:31', 1, '2026-10-10', 'Finished', 1);
```

	participantID	fullName	LastName	email	phoneNumber	password
1	20	Lindiwe	Nkosi	lindiwe.nkosi@gmail.com	0839876543	HASHED_PW_1
2	21	James	Botha	james.botha@gmail.com	0845551234	HASHED_PW_2

Results Messages

	organisserID	organiserName	organiserSurname	email	phoneNumber
1	10	Thabo	Mokoena	thabo.mokoena@raceday.co.za	0821234567
2	11	Sarah	van der Merwe	sarah.vdm@raceday.co.za	0827654321

Links

ST10486463/RaceDay

RaceDay-Prog2B - YouTube

References

Gosselin, Don, et al. *PHP Programming With MySQL*. don gosselin, 2011.

Microsoft. (2026). *ASP.NET Core documentation*. Microsoft Learn.
Microsoft. (2026). *Entity Framework Core documentation*. Microsoft Learn.
Microsoft. (2026). *C# documentation*. Microsoft Learn.
GitHub. (2026). *GitHub documentation: Getting started with GitHub*.
Microsoft. (2026). *.NET documentation*. Microsoft Learn.

W3Schools (2019). *SQL Tutorial*. [online] W3schools.com.